

Unit

1

Lesson
3

Spark The Challenge!

Rolling Forward.

Time Frame: [90 min]

Lesson Objectives:

By the end of this lesson students will be able to:

- Calculate torque using different wheel radii.
- Compare how wheel width impacts grip and turning ability.
- Choose the best wheel for your design based on testing and calculations.

Challenge questions of the lesson

1. How does changing the size of a wheel affect the force needed to turn it?
2. Why might a wider wheel be better on some surfaces but worse on others?
3. What is torque, and why does it matter in movement design?
4. If your wheel is too small, what could go wrong on rough terrain?
5. Which matters more: grip or speed?
6. How can we test wheel options fairly?

SEL Relation

- **SEL Relation: Community Awareness & Problem Solving (Social Awareness & Responsible Decision-Making):** Students will consider real-world problems related to mobility challenges and accessibility for individuals, particularly in diverse environments. They will use their understanding of friction, force, and wheel design to guide their choices for a more effective and user-friendly wheelchair, fostering a sense of empathy and civic responsibility towards designing inclusive solutions.
- **Collaboration and Communication (Relationship Skills):** Students will work effectively together (if in groups) to brainstorm wheel design ideas, share research findings on friction and material properties, troubleshoot technical challenges during lab experiments (e.g., accurate force measurement, torque calculation), and provide constructive feedback on Tinkercad designs. This process develops strong communication and teamwork skills throughout the design and testing process.
- **Innovation and Persistence (Self-Awareness & Self-Management):** Students will face the complex challenge of selecting and designing optimal wheels for specific terrains, requiring them to think creatively to integrate physics principles into their design choices, analyze constraints related to materials and forces, and persist through troubleshooting issues encountered in lab data collection or Tinkercad modeling. This process builds self-awareness of their problem-solving abilities and enhances their self-management when encountering difficulties.
- **Applied Responsibility & Safety (Responsible Decision-Making):** The lesson inherently encourages students to make responsible choices regarding wheel material selection based on performance and safety needs, ensuring accurate measurement of force and radius during the DIY Lab, and practicing safe handling of lab equipment. This fosters a sense of responsibility for their work and the safety of themselves and their peers within the classroom environment and in their design considerations.

Engage



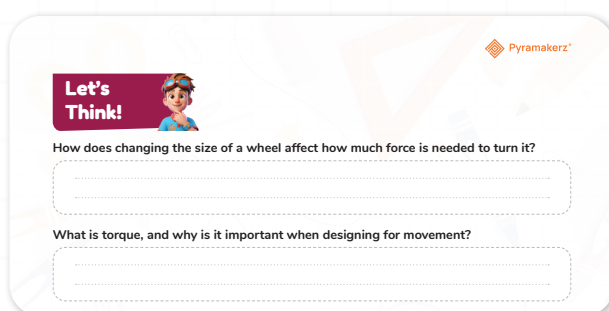
Teacher's Role:

- **Explainer of Physical Principles:** Clarify key concepts such as force, radius, mechanical advantage, and the definition of torque. Explain how these principles apply to the movement and operation of simple machines like wheels and axles.
- **Facilitator of Inquiry:** Ask open-ended questions that prompt students to explore the relationships between different variables.
- **Guide for Real-World Application:** Lead discussions and activities that connect theoretical concepts to practical design challenges. Consider asking: "Can you think of a real-world machine where a large wheel is crucial? What about a small wheel?"

Expected From Students:

- **Active Listening & Conceptual Grasp:** Listen carefully to explanations of force, torque, and wheel mechanics, focusing on understanding definitions and relationships.
- **Prediction & Analytical Thinking:** Hypothesize about the effects of changing variables, such as how altering wheel size might impact the force required to turn it.
- **Problem-Solving & Justification:** Analyze scenarios related to force, torque, grip, and speed, and explain why certain design choices are made in mechanical systems.
- **Application & Design Thinking:** Consider how understanding these physical principles can improve the design and function of everyday objects and machines.

Parts of the book included.



Let's Think!



Objective:

Students will be able to explain and predict how changing wheel size impacts the force needed to turn it, using concepts like mechanical advantage and radius. They will define torque as a rotational force and explain its importance in machine design for movement. Furthermore, students will analyze and justify whether grip or speed is more crucial in various real-world scenarios.

Answers:

- How does changing the size of a wheel affect how much force is needed to turn it? Changing the size of a wheel directly affects the mechanical advantage and the torque produced. A larger wheel (or larger radius at which force is applied) generally requires less force to achieve the same turning effect on its axle, because the force has a longer lever arm. Conversely, a smaller wheel requires more force for the same turning effect. This is due to the principle of torque ($\tau = F \times r$).
- What is torque, and why is it important when designing for movement? Torque is a twisting or turning force that tends to produce rotation. It's the rotational equivalent of linear force. It is important in designing for movement because it determines the ability to start, stop, or change the rotational speed of an object. Enough torque is needed to overcome resistance (like friction or gravity) and to transmit power efficiently in machines like engines, gears, and motors.

Possible Strategies for Implementation:

1. "Force & Fulcrum" Demonstration & Discussion

- How To implement: use simple levers or a wheel-and-axle model. Demonstrate how applying force at different distances from the pivot point (the axle or fulcrum) changes the effort needed to move a load. Ask students to observe and describe what happens when the force is applied closer versus farther from the center. This strategy provides a concrete, hands-on experience that directly illustrates the relationship between force, distance, and turning effect, leading into the concept of torque and addressing the first question.

2. "Machine Match-Up" Challenge

- How To Implement: present students with pictures or descriptions of various machines, such as a bicycle, a car, a crane, a screwdriver, a doorknob, or a skateboard. For each machine, challenge them to identify where large or small wheels/radii are used and explain why that design choice was made in terms of force needed or torque produced. This encourages students to apply their understanding of force, torque, and wheel size to diverse real-world examples, deepening their comprehension and addressing questions 1 and 2.

3. "Grip vs. Speed Scenario Cards" Debate

- **How To implement:** prepare scenario cards with situations like "Designing tires for a monster truck," "Building a robot for a fast delivery service," "Choosing shoes for a rock climber," or "Making a car for a drag race." In small groups, have students discuss for each scenario whether grip or speed (or a balance) is more crucial and why. Encourage them to justify their reasoning using concepts of friction and efficiency. This strategy directly tackles the third question, promoting critical thinking, debate, and the application of concepts to specific contexts, while also fostering persuasive communication and an understanding of design trade-offs.

Explorer's Toolkit



Teacher's Role:

- **Lab Director and Scientific Coach:** You'll guide students through the experimental process, making sure they work safely and collect data accurately.
- **Concept Clarifier:** You'll explain key ideas like torque, force, radius, friction, and stability as they come up during the lab, and help with calculations and observations.
- **Setup Facilitator:** You'll help set up the lab, get materials ready, and show students how to properly measure force with a spring scale and find the radius of a wheel.
- **Critical Thinking Promoter:** You'll encourage students to make precise observations, think deeply about their findings, and connect their lab results to scientific principles when drawing conclusions for the CER chart.

Expected From Students:

- **Active Participation in Lab:** Students will actively take part in the DIY Lab, working with classmates to set up experiments and collect data for both Task A (Torque Trials) and Task B (Friction & Thickness Test).
- **Accurate Data Collection and Calculation:** They'll carefully measure wheel radii and forces, then correctly calculate torque using the formula $\text{Torque} = \text{Force} \times \text{Radius}$.
- **Detailed Observation:** Students are expected to make thorough observations about traction, steadiness, and rolling performance, especially on rough surfaces.
- **Analytical Thinking and Conclusion Drawing:** They'll analyze their data, use evidence to support their conclusions, and clearly explain their reasoning in the CER chart, arguing for the best wheel design for specific uses.

Parts of the book included.



DIY Lab

Objective:

Students will be able to: conduct an experiment to measure and calculate torque for wheels of different sizes; observe and evaluate the performance of wheels with varying thickness on different surfaces; and apply their experimental findings and calculations to justify optimal wheel designs for specific challenges involving movement and overcoming obstacles.

Materials Needed:

- Three wheels of noticeably different sizes (radii)
- Three wheels with different thickness.
- A spring scale
- A ruler or tape measure
- A "rough ground" surface (e.g., a piece of carpet, uneven cardboard, or a mat with small obstacles)
- A flat, smooth surface for comparison
- Laptops
- Access to the internet for the simulation game: <https://www.canva.com/design/DAGsegGkpMQ/Zna3Jpa-Kdv0ZFNPkxOoaA/edit>

Activity Instructions:

Task A: Torque Trials:

Students will:

- Investigate how a wheel's size (radius) influences the torque (twisting force) needed to move over rough ground.
- Test three different-sized wheels on a bumpy surface.
- For each wheel, attach the spring scale and pull it across the surface, measuring the force required to keep it moving at a steady pace.
- Measure the radius of each wheel with the ruler.
- Then, using the formula $\text{Torque} = \text{Force} \times \text{Radius}$, calculate the torque for each trial.



This task will show how wheel size directly impacts the effort to overcome obstacles.

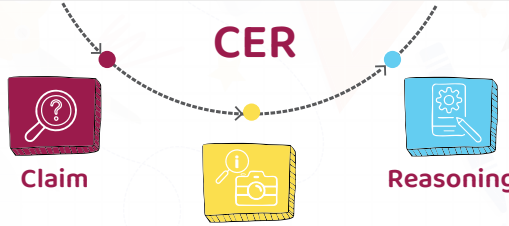
Task B: Friction & Thickness Test:

Students will:

- experiment with three wheels of different thicknesses.
- As you pull each one across a designated surface (both rough and smooth, if possible), you will observe its traction (how well it grips), how steady it feels (its stability), and its overall rolling performance.
- rank them based on these observations, helping you understand how thickness contributes to a wheel's effectiveness.
- After testing let them play the simulation game and write down their observation.
- Calculation and Observations Table:
calculate Torque = Force × Radius

Wheel	Radius (m)	Force (N)	Torque (Nm)	Observations on Surface
1				
2				
3				

- **CER chart:** After completing trials and calculations, consider: which wheel do you think is the best for your design and why? Provide your Claim, Evidence from your data and observations, and your Reasoning that connects the evidence to your claim, explaining the scientific principles involved.



CER

Claim **Evidence** **Reasoning**

What relationship did you observe between the wheel's radius and the amount of torque needed to move it over the bumpy surface?

Which wheel size (radius) do you claim required the least/most torque to overcome obstacles?

The smaller the wheel's radius, the less torque is required to move it over the bumpy surface.

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Describe the observable performance of each wheel as it moved over the bumpy surface during the test.

As the wheels with smaller radii were tested across the bumpy surface, they experienced less resistance and were easier to move. The larger wheels, on the other hand, required more force to overcome the rough texture and obstacles on the surface, indicating that they required more torque to move.

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How does your evidence support your claim about the most/least effective wheel size for torque?

According to the formula $\text{Torque} = \text{Force} \times \text{Radius}$, $\text{Torque} = \text{Force} \times \text{Radius}$, a smaller radius results in lower torque because the force required is distributed across a smaller distance, making it easier to overcome resistance. Thus, the smaller the wheel's radius, the less force and torque are needed to move it over the bumpy surface, making it more efficient for this type of movement.

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Possible Strategies for Implementation:

1. Pre-Lab Discussion and Hypothesis Formulation

- **How to Implement:**
Before beginning the DIY Lab, engage students in a brief discussion to activate prior knowledge about wheels, forces, and motion. Prompt them to formulate hypotheses for both Task A and Task B. For instance, ask them to predict whether larger or smaller wheels will require more torque on a rough surface, and how wheel thickness might influence stability or traction.

2. Guided Measurement and Calculation Practice

- **How to Implement:**
During the lab, provide clear, step-by-step guidance on how to accurately measure the radius of the wheels and how to use the spring scale to obtain consistent force readings. Model one or two torque calculations explicitly, ensuring students understand the units and the application of the formula.

3. Data Analysis and Conclusion Synthesis Workshop

- **How to Implement:**
After students complete their experiments and fill in the table, dedicate time to a structured data analysis session. Guide them in interpreting their torque values and comparing their qualitative observations across different wheels. Provide support for constructing their CER charts, emphasizing the importance of using specific numerical evidence from their tables and detailed descriptions from their observations to support their claims. Encourage them to articulate the scientific reasoning that links their evidence to their conclusions about optimal wheel design.

Explain



Teacher's Role:

- **Concept Illuminator:** You'll use the video to clearly explain how wheel size, force, and torque are interconnected. You'll highlight how these concepts are applied in real-world machines, making the abstract visible.
- **Visual Learning Facilitator:** You'll guide students through the video content, pausing to discuss key demonstrations and visuals that illustrate the impact of wheel size on torque and movement.
- **Discussion Provoker:** You'll pose questions before, during, and after the video to stimulate critical thinking about how different wheel designs affect machine performance and efficiency.
- **Application Connector:** You'll help students bridge the gap between the video's theoretical explanations and practical engineering challenges, encouraging them to consider the implications for designing real-world mobility solutions.

Expected from Students:

- **Active Viewing & Note-Taking:** Students will watch the video attentively, focusing on the demonstrations and explanations regarding wheel size, force, and torque. They will take notes on key concepts and observations.
- **Conceptual Understanding:** They will strive to understand the relationship between wheel size, the force needed to turn it, and the resulting torque, as presented in the video.
- **Critical Analysis:** Students will analyze the video's examples to grasp how different wheel characteristics affect machine performance in various scenarios.
- **Engaged Discussion:** They will participate actively in discussions, sharing their interpretations of the video, asking clarifying questions, and connecting the video's content to broader engineering principles.

Parts of the book included.



Screen Lessons

Why Wheel Size Matters in Engineering?

Click on the link to visualize how size and surface area affect real machines.

<https://www.youtube.com/watch?v=W-3GOVKRlpo>



Screen Lessons

Objective:

Students will be able to visualize and explain how changing wheel size affects the amount of force needed to turn it and the resulting torque, as demonstrated in the video, thereby deepening their understanding of mechanical advantage in engineering design.

Video link: <https://www.youtube.com/watch?v=W-3GOVKRlpo>



Possible Strategies for implementation:

1. Pre-Watching: Set a Purpose

- **How to Implement:** Before playing the video, introduce the topic by asking students: "We're going to watch a video about how wheel size affects how much effort it takes to move something. What do you already know about big wheels versus small wheels on things like bikes or cars? What do you predict the video will show about bigger tires and torque?" This sets a clear viewing purpose and activates prior knowledge.

2. During Watching: Active Engagement

- **How to Implement:** Pause the video at key demonstration points (e.g., when different sized tires are being tested, or when torque is explained visually). Ask guiding questions like: "What did you observe in that demonstration? How does the force change when the wheel size changes?" or "How is the video explaining what torque is here?" Encourage students to jot down their observations and questions.

3. Post-Watching: Discussion and Reflection

- How to Implement: After the video, facilitate a class discussion. Ask students to summarize the main takeaways regarding wheel size and torque. Pose questions such as: "How does this video help you understand why some vehicles have very large wheels and others have small ones?" or "Can you give an example of a machine where a large wheel would be essential, based on what you learned about torque?"



Dig Deeper

Objective:

Students will develop a deeper understanding of how friction, force, and specific wheel design elements impact the performance of wheelchairs on various terrains. By actively engaging with the article "Wheelchair Wheels: What You Need To Know For Performance Purposes – Reckless Wheelchairs," they will practice critical thinking, information gathering, and communication skills as they analyze how different wheel features are engineered to overcome environmental challenges, revise initial assumptions, and articulate new insights about designing for optimal mobility.

Article link: https://recklesswheelchairs.com/pages/wheelchair-wheels-what-you-need-to-know-for-performance-purposes?srsltid=AfmBOooLBZpYV47lA_Xik9vv9-JD7extlp1VXGXW1hU2nlwqY706lmHO



Possible Strategies for implementation:

1. Pre-Reading: Activating Prior Knowledge & Setting Purpose

Strategy: KWL Chart - "Wheel Performance Ponder!"

- How to Implement:
 - Introduce the article topic: "Today, we're going to read about the incredible engineering behind wheelchair wheels! Just like cars need different tires for different roads, wheelchairs need specific wheels for different activities and terrains. Before we start, let's think about what we already know and what we want to find out about how wheel design affects performance."
 - Distribute a KWL chart (Know, Want to Know, Learned) or draw one on the board.
 - "K" (Know): Ask students, "What do you already know about wheels, friction, or how different surfaces affect movement?" Have them individually fill in the "K" column with their existing knowledge (e.g., "Smooth tires slip on ice," "Mountain bike tires have big bumps for grip").
 - "W" (Want to Know): Next, ask, "What do you want to learn or wonder about how wheelchair wheels are designed for performance, or how friction and force play a role?" Have them fill in the "W" column.
 - Explain that reading the article will help them fill in the "L" column later.
 - Please: print them the following chart: <https://g.co/gemini/share/9a5ceefdf570>

2. During Reading: Focused Information Gathering

Strategy: Guided Reading with Highlighting/Note-Taking - "Wheel Design Detectives!"

- How to Implement:
 - Instruct students to read the article ("Wheelchair Wheels: What You Need To Know For Performance Purposes – Reckless Wheelchairs") either individually, in pairs, or as a class.

K- What I know	W- What I want to know	L- What I Learned

- Provide specific prompts to guide their reading. Examples:
 1. "As you read, try to find at least three different features of a wheelchair wheel that affect its performance (e.g., size, material, tread)."
 2. "For each feature, look for how it influences friction or force."
 3. "Look for examples of what kind of terrain or activity each wheel design is best suited for."
 4. "Pay attention to any words that describe how wheels gain grip or reduce resistance."
- Encourage students to highlight or jot down brief notes/keywords related to these prompts directly on a paper or in their digital document.

3. Post-Reading: Consolidating Learning & Reflection

Strategy: KWL Chart Completion & Discussion - "Wheel Design Experts Share!"

How to Implement:

- Bring students back together.
- "L" (Learned): Ask students to fill in the "L" column of their KWL chart with new information they learned from the article, paying special attention to answering questions from their "W" column.
- Share and Discuss: Facilitate a whole-class discussion where students share what they wrote in their "L" column. Ask for examples of different wheel designs and how they manage friction and force for performance.

- Reflect: Ask concluding questions: "What was the most surprising thing you learned about how wheelchair wheels are designed?" or "If you were designing a wheelchair for someone who loved to go off-roading, what kind of wheels would you recommend and why, based on this article?"

Elaborate and Evaluate



Teacher's Role:

- **Assessor of Understanding:** You will use these questions to gauge students' comprehension of key concepts such as torque, force, radius, friction, and the design process related to wheel mechanics.
- **Concept Reinforcer:** After students complete the questions, you will facilitate a review session, explaining the correct answers and clarifying any misconceptions about the relationships between force, radius, and torque, as well as the critical role of friction.
- **Guide for Application:** You will connect these questions back to the practical aspects of engineering and design, emphasizing how theoretical knowledge translates into real-world decisions, particularly in selecting components based on data.
- **Feedback Provider:** You will offer constructive feedback on student responses, helping them to solidify their grasp of the physics principles and their application in problem-solving.

Expected From Students:

- **Application of Knowledge:** Students will apply their understanding of torque calculation, the relationship between force and radius, the role of friction, and the steps of the design process to answer the given questions.
- **Critical Thinking:** They will carefully read and analyze each question, evaluating the provided options to select the most accurate answer based on scientific principles.
- **Demonstration of Learning:** Students will demonstrate their individual comprehension of the core concepts covered in the lesson by accurately responding to the multiple-choice questions.
- **Reflective Engagement:** They will participate in post-assessment discussions, reflecting on their answers and clarifying any areas of confusion.

Parts of the book included.



Apprentice

Let's put our learning to the test!

https://www.canva.com/design/DAGsegGkpMQ/Zna3Jpa-Kdv0ZENPKxOoaA/edit?utm_content=DAGsegGkpMQ&utm_campaign=designshare&utm_medium=link2&utm_source=sharebutton



Figure 10: Wheel Physics Quiz Interface.



Figure 11: Close-Up View of Wheelchair Wheels.



Apprentice

Objective:

Students will demonstrate their foundational understanding of torque, friction by correctly answering multiple-choice questions related to wheel mechanics and data-driven decision-making.

Answers:

What happens to torque when the wheel radius increases, assuming the force stays the same?

- C. Torque increases

Explanation: Torque (τ) is calculated as Force (F) multiplied by Radius (r) ($\tau = F \times r$). If the force remains constant, increasing the radius will directly increase the resulting torque.

Which factor is most responsible for how easily a wheel rolls on a surface?

- D. Friction between the wheel and the surface

Explanation: While other factors can play a role, the amount of friction (specifically rolling resistance) between the wheel and the surface is the primary determinant of how easily it rolls. Less rolling friction means it rolls more easily.

Which formula do we use to calculate torque in this lesson?

- C. Force $\times$ Radius

Explanation: This is the standard formula for calculating torque, where Force is the applied force and Radius is the perpendicular distance from the axis of rotation to the line of action of the force.

After testing and calculating torque, what's the next step in the design process?

- D. Use your data to select the best wheels for your prototype

Explanation: The engineering design process is iterative and data-driven. After testing and collecting data, the crucial next step is to analyze that data to make informed decisions and select components that best meet the design criteria for the prototype.

Possible Strategies for Implementation:

Strategy: Diagnostic Check-in/Implementation:

- How to Implement:
Present these questions as a quick quiz at the beginning of a class or after a specific concept has been taught. This helps you quickly identify areas where students might need more instruction or clarification before moving on.

Strategy: Collaborative Problem-Solving

- How to Implement:
Have students work in small groups to discuss and answer the questions. Encourage them to justify their choices to each other, referencing notes or previous lab findings. This promotes peer learning and deeper understanding through discussion.

Strategy: Exit Ticket Assessment

- **How to Implement:**
Use these questions as an exit ticket at the end of a lesson. Students answer individually before leaving, providing you with a snapshot of their individual comprehension that can inform future lesson planning.

Strategy: Interactive Review Game

- **How to Implement:**
Transform these questions into an interactive game format (e.g., using online quiz tools like Kahoot or Quizizz, or a simple classroom game). This can make the assessment more engaging and provide immediate feedback to students.



Builder

Objective:

Students will design a wheelchair in Tinkercad, focusing on selecting and justifying the most suitable wheels for a specific user and environment.

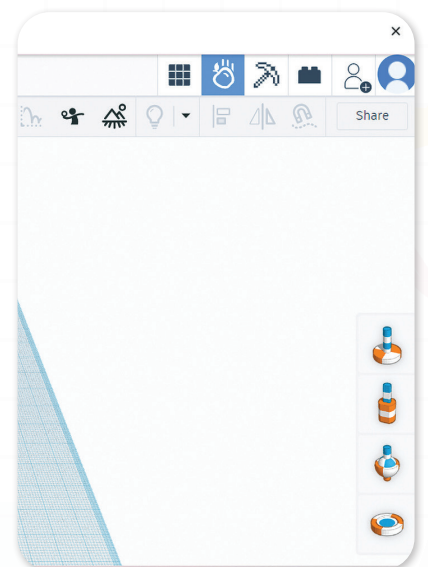
Materials Needed:

- laptops with internet access
- Access to Tinkercad (<https://www.tinkercad.com/>)
- Optional: Sketching paper and pencils for initial design ideas



Activity Instructions:

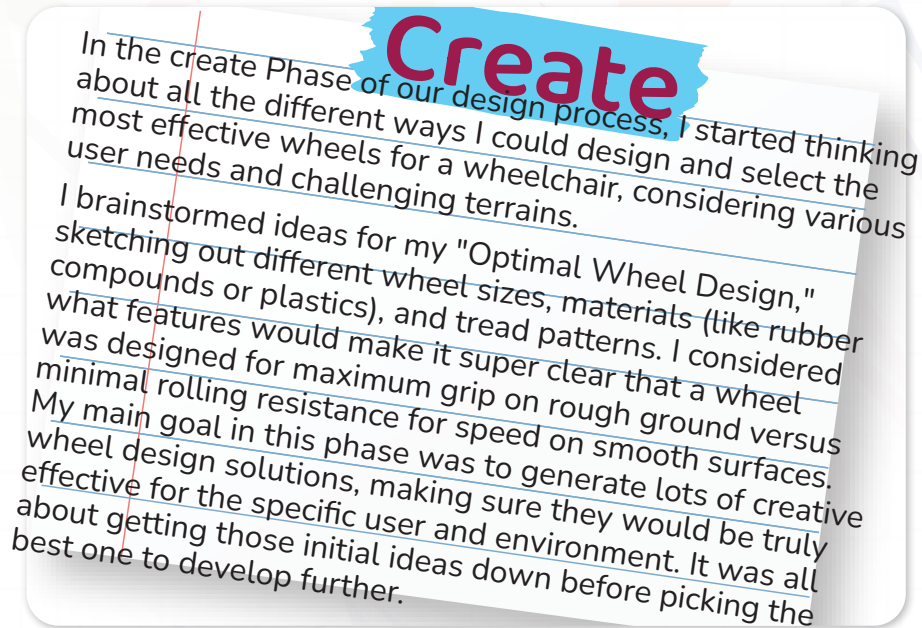
- **Choose a User & Environment:** Begin by deciding who will use your wheelchair and where they will primarily use it
- **Research Wheels:** Investigate different wheel types, considering their size, material (e.g., rubber, plastic, pneumatic), and tread patterns. Research how these characteristics affect suitability for various conditions, paying attention to concepts like friction, rolling resistance, and shock absorption.
- **Design in Tinkercad:**
 - Build a simple wheelchair frame. Focus on the basic structure needed to support the wheels and a seat.
 - Design and attach the main wheels (large rear wheels) and caster wheels (small front wheels) that best fit your chosen user and environment. Pay attention to their scale relative to the frame.
- **Justify Your Choices:** Be prepared to explain why your chosen wheels (size, material, tread) are ideal for your specific user and environment. Your justification should reference the research you conducted on friction, force, and wheel performance.
- Ask them to use sim lab to simulate the movement of the wheelchair



Tinkercad Robotics for School Wheelchair Simulator: <https://www.tinkercad.com/projects/Tinkercad-Robotics-for-School-Wheelchair-Simulator>



It's time to move to the Create part and add this.



Now I Can

Objective:

Students will reflect on their learning journey to define torque, force, and motion, and calculate torque; explain how friction affects movement and compare the impact of wheel width; compare how different wheel sizes and forces influence torque; and ultimately choose the best wheel for their design, justifying their selection based on research and calculations.

Activity Instructions:

Step 1: Revisit Your Work and Data

- Review your DIY Lab results: Look back at your "Wheel Testing & Torque Calculation Challenge" data table (including Radius, Force, Torque, and Observations) and your CER chart.
- Re-examine your KWL Chart: Read through your "KNOW," "WONDER," and especially your "LEARNED" sections from the sensor and wheel design articles.
- Recall your Tinkercad Wheelchair Challenge: Think about the design choices you made for your wheelchair in Tinkercad, particularly concerning the main wheels and caster wheels.

Step 2: Student Self-Reflection

- Individually, take a few minutes to think about your journey through these activities. Consider the following:
 - How well can you now define torque, force, and motion? Can you confidently calculate torque?

- How has your understanding of friction evolved, and how does wheel width specifically impact movement?
- What did you learn about how different wheel sizes and applied forces affect the resulting torque?
- Based on all your research and calculations, how confident are you in choosing the best wheel for a specific design challenge? What data or insights would you use to justify your choice?

Step 3: Small Group Discussion

- Form small groups and share your reflections. Discuss the following:
 - Compare your answers to the reflection questions. Did you all have similar insights, or were there differences?
 - Share one key takeaway about torque, friction, or wheel design that surprised you or that you found particularly important.
 - Discuss how the DIY Lab, the KWL Chart, and the Tinkercad design each contributed to your understanding of these concepts.
 - As a group, identify one challenge you faced during these activities and how you overcame it (or how you might overcome it in the future).

Relation to Standards

1. Next Generation Science Standards (NGSS):

- **MS-PS2-1:** Apply scientific principles to explain the behavior of objects.
 - (Students are applying principles of force, friction, and torque to understand how wheels move and interact with surfaces.)
- **MS-ETS1-1:** Define the criteria and constraints of a design problem with sufficient precision to ensure a successful solution, taking into account relevant scientific principles.
 - (Students are defining the problem of designing a wheelchair for a specific user and environment, considering performance criteria and wheel characteristics as constraints.)
- **MS-ETS1-2:** Evaluate competing design solutions using a systematic process to determine how well they meet the criteria and constraints of the problem.
 - (Students are evaluating different wheel types and their suitability, and justifying their choices based on research and lab data.)
- **MS-ETS1-3:** Analyze data from tests to determine similarities and differences among several design solutions to identify the best characteristics of each that can be combined into a new solution.
 - (Students are analyzing data from their DIY Lab experiments on torque, friction, and thickness to inform their wheelchair wheel selection.)

- **MS-ETS1-4:** Develop a model to generate data for iterative testing and modification of a proposed object, tool, or process such that an optimal design can be achieved.
 - (Students are using Tinkercad to develop a model of a wheelchair, applying their understanding to create an optimal design for specific conditions.)
 - (Students apply the torque formula ($\tau = F \times r$) and implicitly understand how changing variables affects the outcome, which relates to rearranging formulas.)

2. Common Core State Standards (CCSS) - English Language Arts (Speaking and Listening):

- **SL.9-10.1:** Initiate and participate effectively in a range of collaborative discussions (one-on-one, in groups, and teacher-led) with diverse partners on grades 9–10 topics, texts, and issues, building on others' ideas and expressing their own clearly and persuasively.
 - (Students engage in small group discussions, sharing reflections, comparing insights, and justifying their design choices.)
- **SL.9-10.4:** Present information, findings, and supporting evidence clearly, concisely, and logically such that listeners can follow the line of reasoning and the organization, development, substance, and style are appropriate to purpose, audience, and task.
 - (Students justify their wheel choices for the Tinkercad design and articulate their learning in reflections, using evidence from research and lab activities.)

3. Common Core State Standards (CCSS) - Mathematics (Number and Quantity/Algebra):

- **HS.N-Q.A.1:** Use units as a way to understand problems and to guide the solution of multi-step problems; choose and interpret units consistently in formulas; choose and interpret the scale and the origin in graphs and data displays.
 - (Students use units like meters (m), Newtons (N), and Newton-meters (Nm) consistently in their torque calculations and data tables.)
- **HS.A-CED.A.4:** Rearrange formulas to highlight a quantity of interest, using the same reasoning as in solving equations.